Theme and chassis layout

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Make a dimensioned layout that supports the circuit and drivetrain.

SUCCESS CRITERIA - I can...

1. Label component locations and clearances.
2. Use dimensions to check that parts fit.
3. Explain a structural choice with evidence.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A motor uses 4 W for 6 s. Find electrical energy and name a possible non-mechanical output.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: available length = chassis length - components - clearances.
Use measured dimensions for a top view. Keep axles parallel and controls accessible. Get approval before tools/cuts.

Worked reasoning: A 20 cm chassis reserves 6 cm for the battery, 5 cm for the motor and 2 cm total for end clearances. The remaining 7 cm must cover other components and gaps.

Your turn: Draw a dimensioned top view showing these regions. Label any assumed dimensions rather than pretending they were measured.

Marzano check - circle: ① ② ③

03 Look and explain

Draw a dimensioned top view of your chassis. Label motor, battery, switch, axle lines and wheel clearance. Mark the intended forward direction.



Top view: the large rectangle is the chassis; small rectangles are wheels; the two crossbars are axles.

Marzano check - circle: ① ② ③

04 Build the solution

A supplied chassis is 18 cm long. The battery needs 6 cm and the motor zone needs 5 cm along its length. Find remaining length before allowing for gaps. Then mark the actual gaps on your own plan.

Marzano check - circle: ① ② ③

Theme and chassis layout

antocciphysics

Continue down each column.

05 Use the idea

Your theme adds a heavy decoration far to one side. Predict a possible effect and propose a way to test it without claiming the effect is certain.

Marzano check - circle: 1 2 3

06 Choose a method

A total current of 16 A splits between two identical parallel branches. Find each branch current.

Marzano check - circle: 1 2 3

07 Exit check

Name the measurement you will verify before the first cut and record the teacher checkpoint.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Return to the worked example in problem 2. Fit the same components on an 18 cm chassis. Propose two layouts and identify a measurement needed before cutting.

Marzano check - circle: 1 2 3